A Note on "FLRW Cosmology in Generic Gravity Theories": Two Non-Riemannian Examples

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Abstract

In [1], the authors proved that, for the Friedmann–Lemaître–Robertson–Walker (FLRW) metric, the field equations of a broad class of metric gravity theories reduce to the perfect-fluid form. In this note, we examine whether the same structural behavior is visible in two non-Riemannian settings: Yano–Schrödinger cosmology [2] and Einstein–Cartan theory [3]. In both examples, after imposing the isotropy and homogeneity assumptions appropriate to an FLRW background, the geometric correction terms can be absorbed into a perfect-fluid form. These examples illustrate how the perfect-fluid structure can persist in the presence of non-metricity or torsion.

1 Introduction

The study of generic gravity theories in the context of FLRW cosmology often relies on the restrictive symmetries of the metric. It has been established that, for theories constructed from the metric and the Riemann tensor, the field equations reduce to the perfect-fluid form [1]. A natural question is whether analogous behavior persists when one allows non-Riemannian features such as torsion or non-metricity.

In this note, we consider two examples in which this question can be examined explicitly. The first is Yano–Schrödinger cosmology [2], where the geometry has vanishing torsion but non-vanishing non-metricity. The second is Einstein–Cartan theory [3], where the connection is metric-compatible but possesses non-vanishing torsion. In both examples, we show that the non-Riemannian corrections are compatible with the perfect-fluid structure once the appropriate isotropy assumptions are imposed.

Throughout, we use the metric signature $(-,+,+,+)$.

2 The Yano–Schrödinger Case

We work with the spatially flat FLRW metric

$$ds^2 = -dt^2 + a^2(t)\delta_{ij}dx^i dx^j. \tag{1}$$

We write

$$H = \frac{\dot{a}}{a}$$

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for the Hubble function.

Let

$$u^\mu = (1, 0, 0, 0)$$

be the comoving four-velocity, and define the spatial projector by

$$h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu.$$

In Yano–Schrödinger cosmology, the geometry is characterized by a length-preserving non-metricity vector π_μ. The generalized Einstein field equations in this framework take the form [2]

$$G_{\mu\nu} + \mathcal{H}_{\mu\nu}(\pi) = 8\pi T_{\mu\nu}, \tag{2}$$

where $G_{\mu\nu}$ is the standard Einstein tensor and $\mathcal{H}_{\mu\nu}(\pi)$ represents the geometric modification due to non-metricity:

$$\mathcal{H}_{\mu\nu}(\pi) = -\frac{5}{4} g_{\mu\nu} \nabla_\alpha \pi^\alpha + \frac{1}{2} \nabla_{(\mu} \pi_{\nu)} + \frac{5}{8} g_{\mu\nu} \pi^\alpha \pi_\alpha - \frac{1}{4} \pi_\nu \pi_\mu. \tag{3}$$

To investigate the perfect-fluid structure, we impose the symmetries of the FLRW spacetime. Spatial isotropy and homogeneity require the non-metricity vector to be orthogonal to the spatial hypersurfaces. Thus, in comoving coordinates, it takes the form [2]

$$\pi_\mu = (-\Pi(t), 0, 0, 0). \tag{4}$$

Substituting this symmetry-constrained vector into the geometric correction tensor $\mathcal{H}_{\mu\nu}$, the off-diagonal components vanish. The remaining diagonal contributions can be represented by an effective geometric energy density and pressure. We denote these correction terms by ρ_{geom} and p_{geom}. They are given by

$$\rho_{\text{geom}} = \frac{1}{32\pi} \left(3\dot{\Pi} + 15 H \Pi - \frac{3}{2} \Pi^2 \right), \tag{5}$$

and

$$p_{\text{geom}} = \frac{1}{32\pi} \left(-5\dot{\Pi} - 13 H \Pi + \frac{5}{2} \Pi^2 \right). \tag{6}$$

Consequently, the modified field equations may be written in the standard perfect-fluid form

$$G_{\mu\nu} = 8\pi \left[(\rho_m + \rho_{\text{geom}}) u_\mu u_\nu + (p_m + p_{\text{geom}}) h_{\mu\nu} \right]. \tag{7}$$

Thus, the non-metric geometric terms are algebraically indistinguishable from an additional perfect-fluid source once the non-metricity vector is constrained by the background symmetries.

3 The Einstein–Cartan Case

In the Einstein–Cartan framework, the geometry remains metric-compatible, but the connection possesses a non-vanishing torsion tensor $T^\lambda_{\mu\nu}$ sourced by the intrinsic spin of matter. The generalized field equations may be written in the form [3]

$$G_{\mu\nu} - \Lambda g_{\mu\nu} = \kappa \left(T_{\mu\nu} + \theta_{\mu\nu} \right), \tag{8}$$

where Λ is the cosmological constant, κ is the gravitational coupling constant, $T_{\mu\nu}$ is the stress-energy tensor, and $\theta_{\mu\nu}$ represents the correction terms arising from the spin–torsion interaction.

For simplicity, we set

$$\Lambda = 0$$

in what follows.

We model the matter source as a Weyssenhoff fluid, which generalizes a perfect fluid by allowing a non-vanishing microscopic spin density $S_{\mu\nu}$. Consistent with an isotropic cosmological background, we assume that the macroscopic average of the spin orientation vanishes,

$$\langle S_{\mu\nu} \rangle = 0,$$

while the spin-squared quantity

$$S^2 := \frac{1}{2} \langle S_{\mu\nu} S^{\mu\nu} \rangle$$

remains non-zero.

In the isotropic limit $p_r = p_t = p$, the stress-energy expression used in [3] reduces to

$$G_{\mu\nu} = \kappa \left(\rho + p - \frac{\kappa}{2} S^2 \right) u_\mu u_\nu + \kappa \left(p - \frac{\kappa}{4} S^2 \right) g_{\mu\nu}. \tag{9}$$

The spin–torsion corrections can therefore be absorbed into redefined energy density and pressure:

$$p_{\text{eff}} = p - \frac{\kappa}{4} S^2, \tag{10}$$

$$\rho_{\text{eff}} = \rho - \frac{\kappa}{4} S^2. \tag{11}$$

Indeed,

$$\rho_{\text{eff}} + p_{\text{eff}} = \rho + p - \frac{\kappa}{2} S^2, \tag{12}$$

so that the field equations become

$$G_{\mu\nu} = \kappa \left[(\rho_{\text{eff}} + p_{\text{eff}}) u_\mu u_\nu + p_{\text{eff}} g_{\mu\nu} \right]. \tag{13}$$

Hence, the torsion-induced correction terms preserve the algebraic perfect-fluid form and can be interpreted as shifts in the effective energy density and pressure.

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References

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